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DEPARTMENT OF
AGRICULTURE, FOOD
ENVIRONMENT AND FORESTRY



HR EXCELLENCE IN RESEARCH

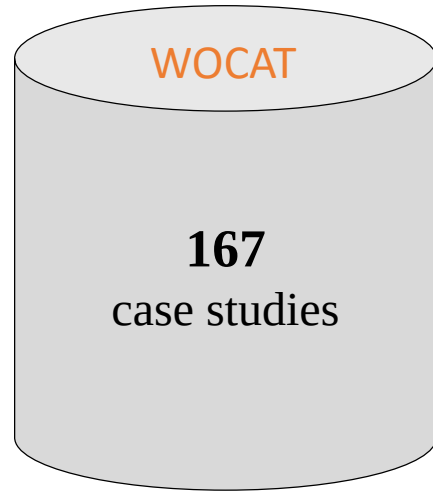
Estimating the global potential of water harvesting from successful cases

Luigi Piemontese, PhD

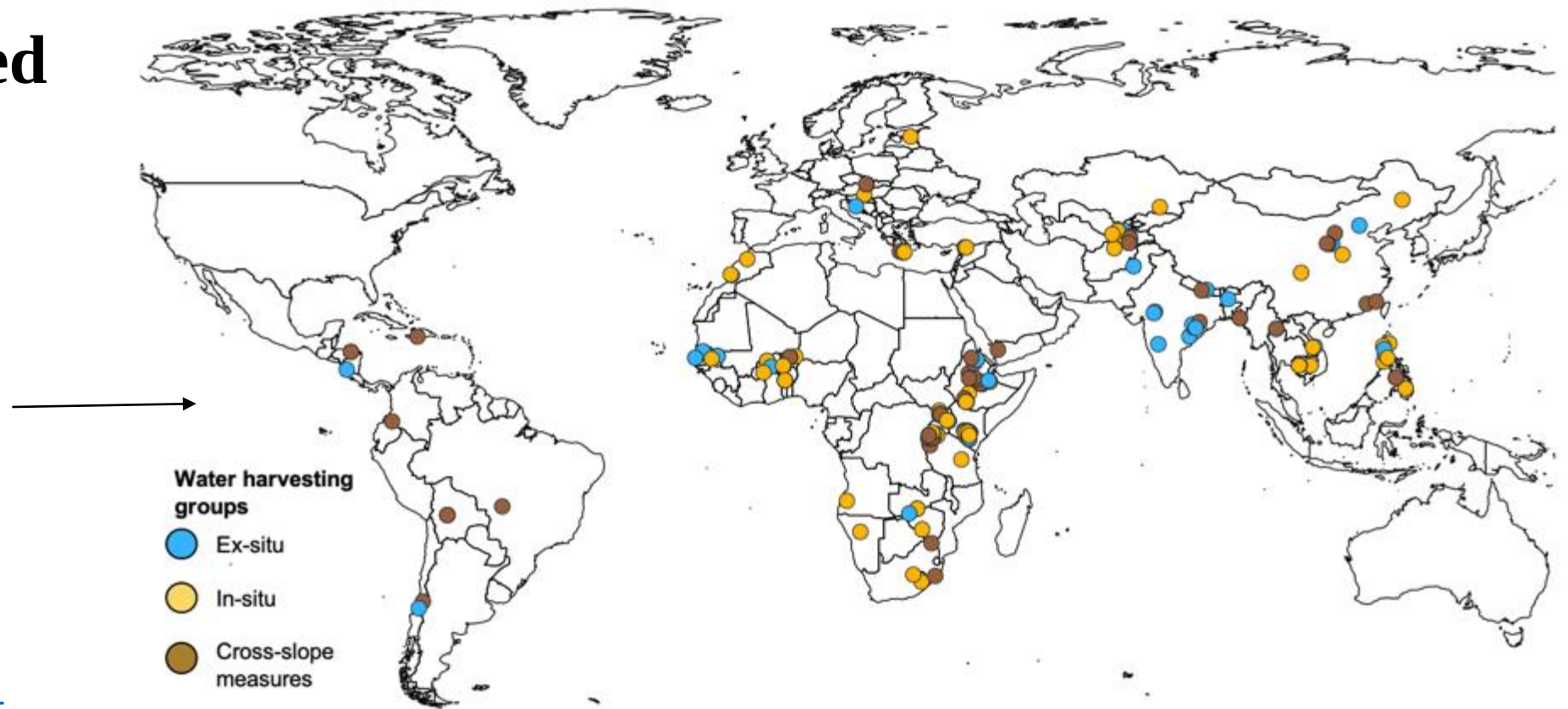
Department of Agriculture, Food, Environment
and Forestry (DAGRI), University of Florence¹



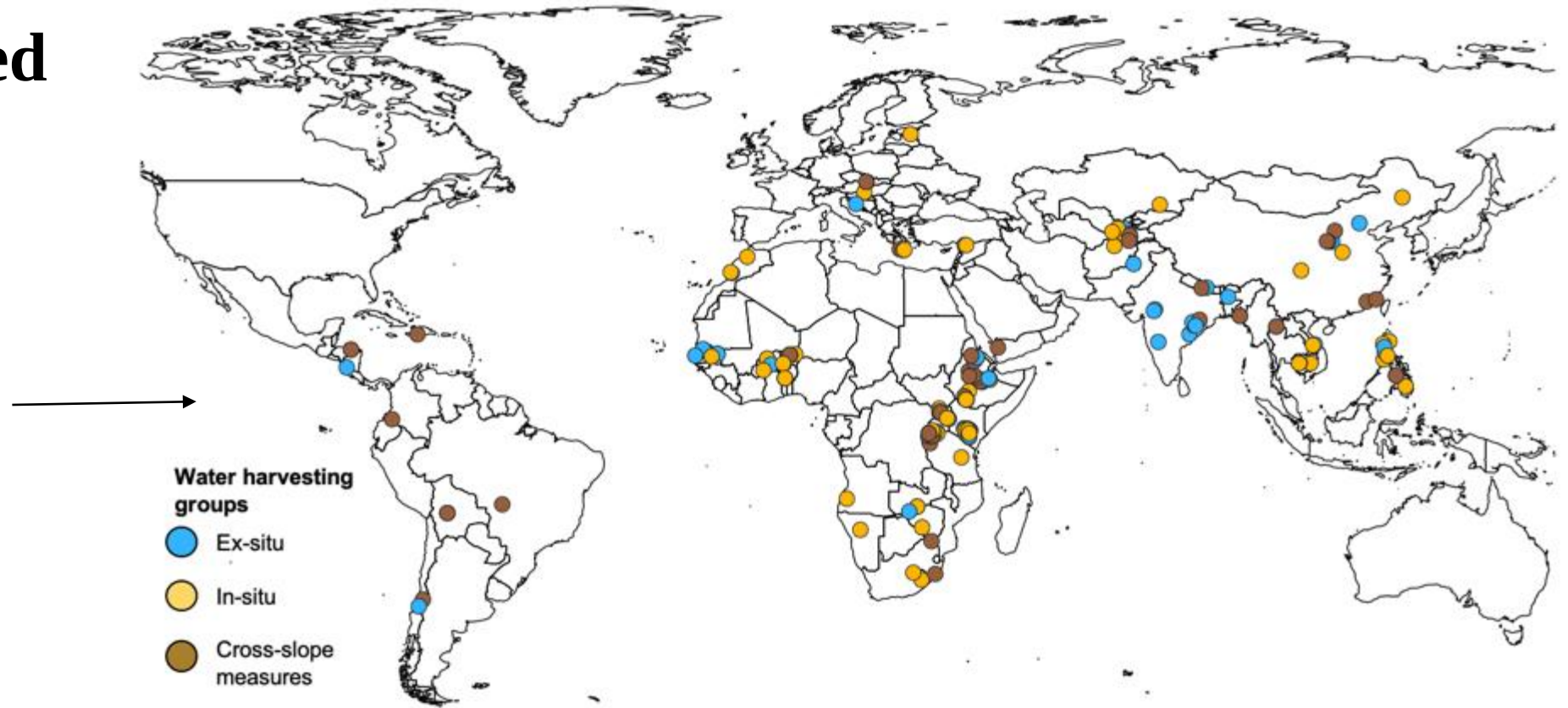
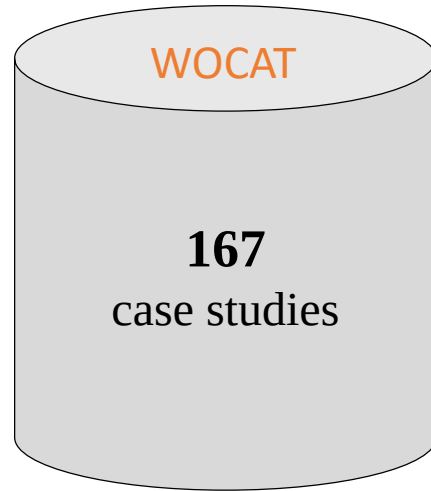
Evidence-based



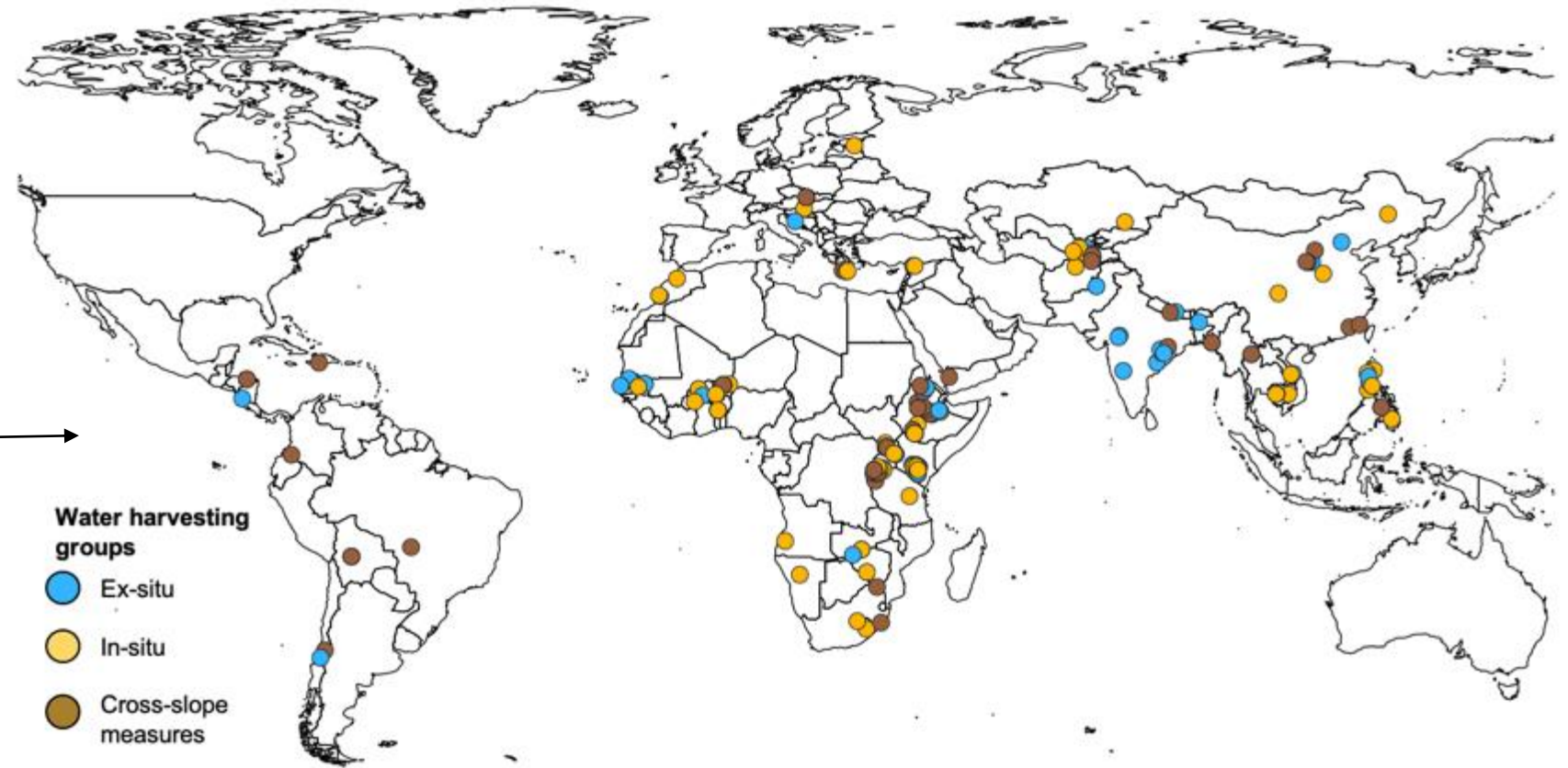
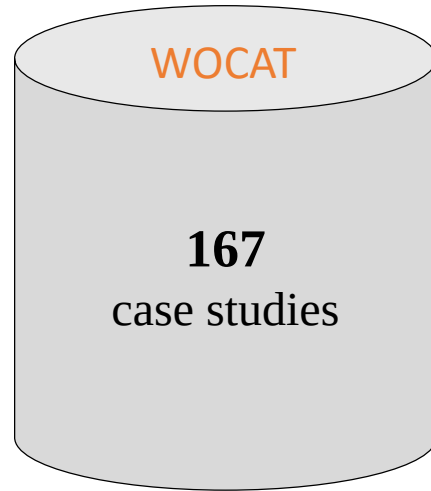
<https://qcat.wocat.net>



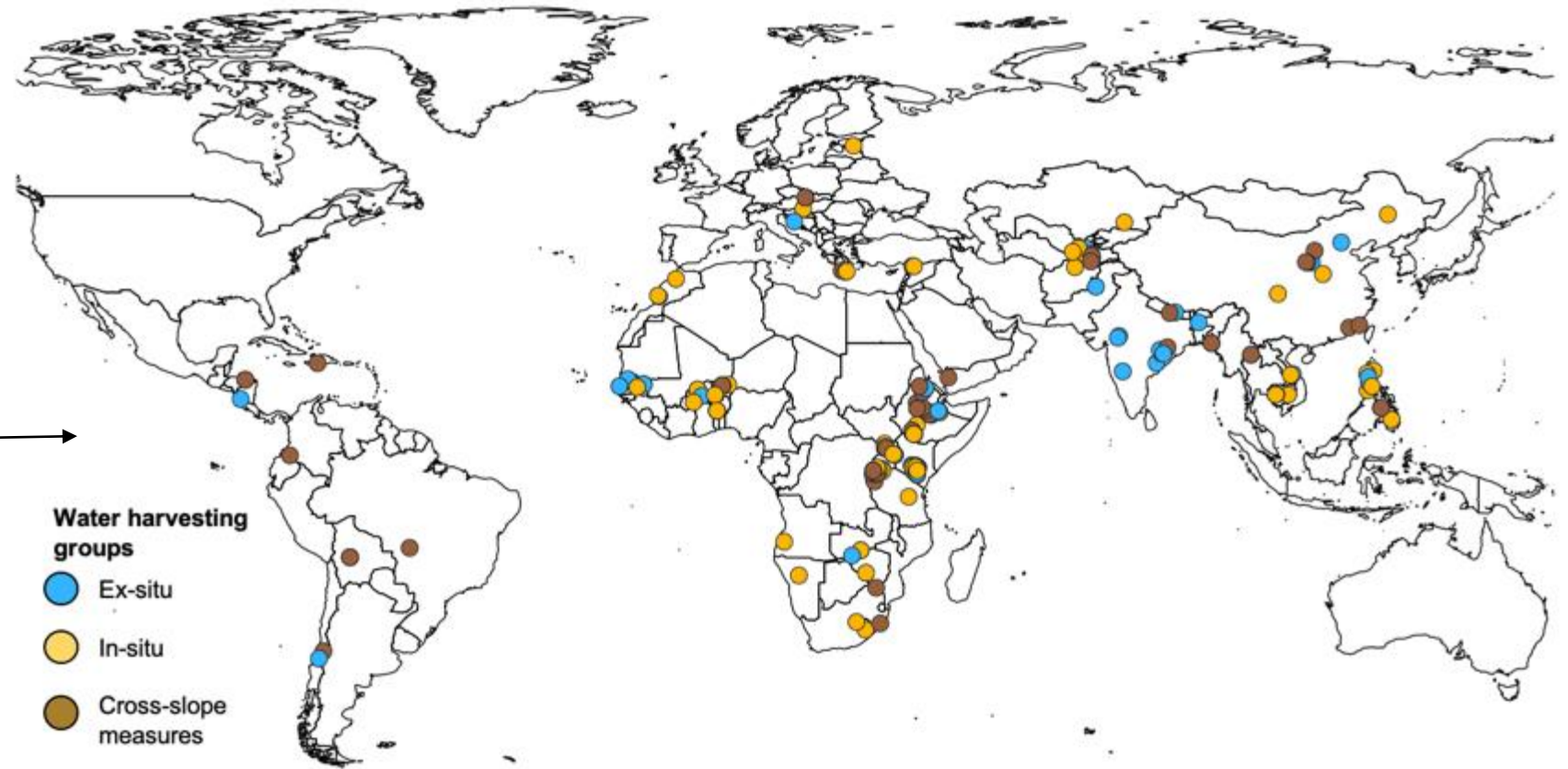
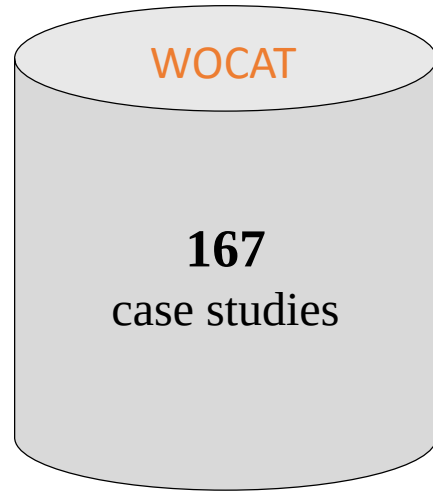
Evidence-based



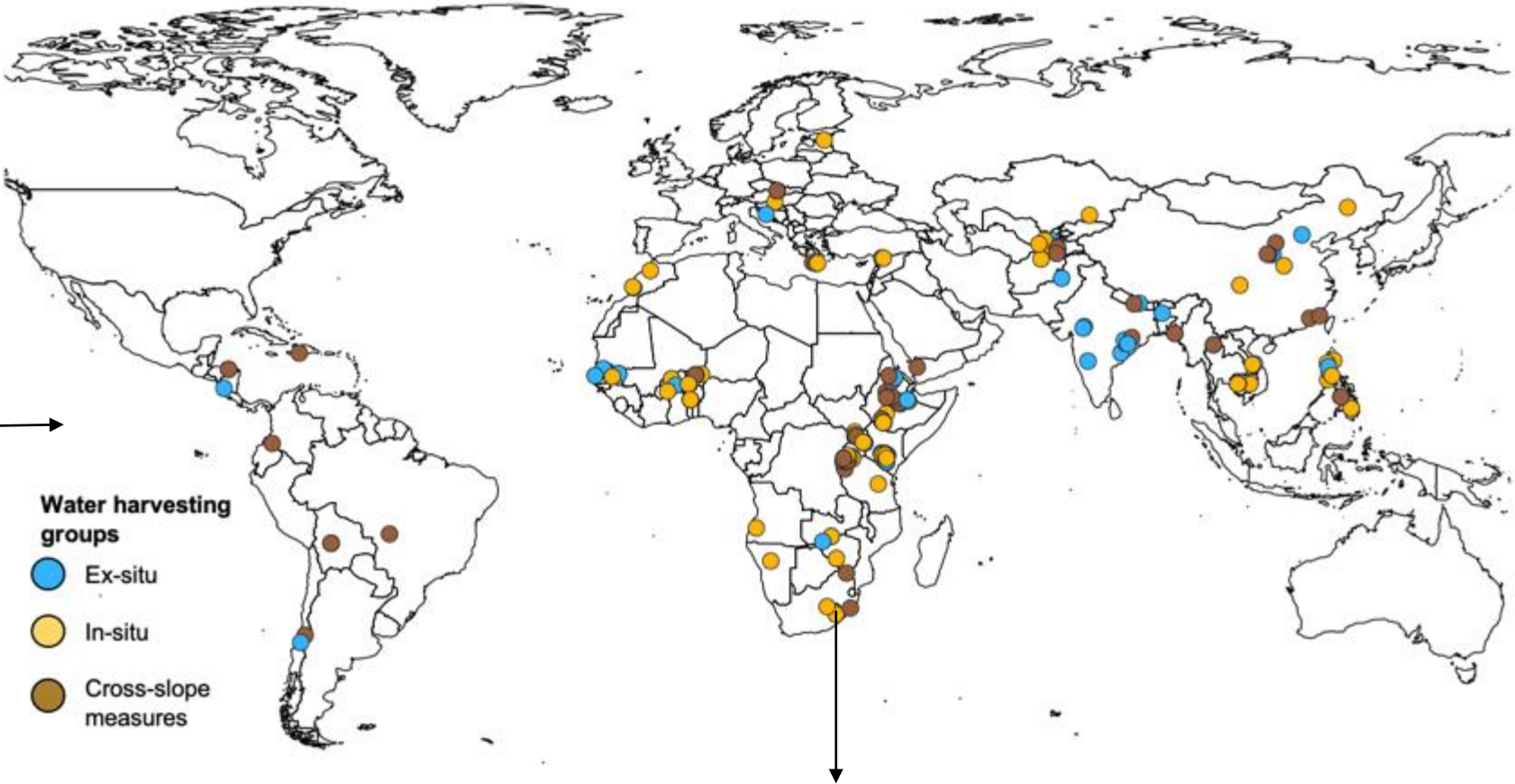
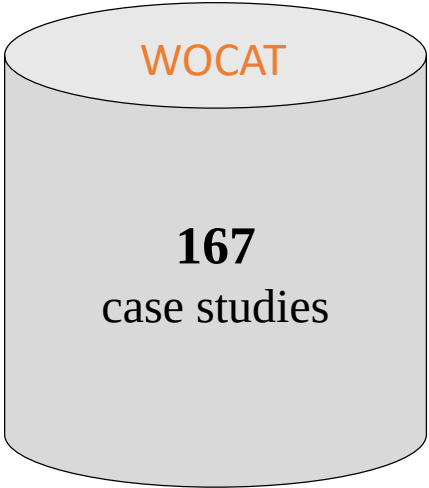
Evidence-based



Evidence-based



Evidence-based



Water harvesting groups

- Ex-situ
- In-situ
- Cross-slope measures

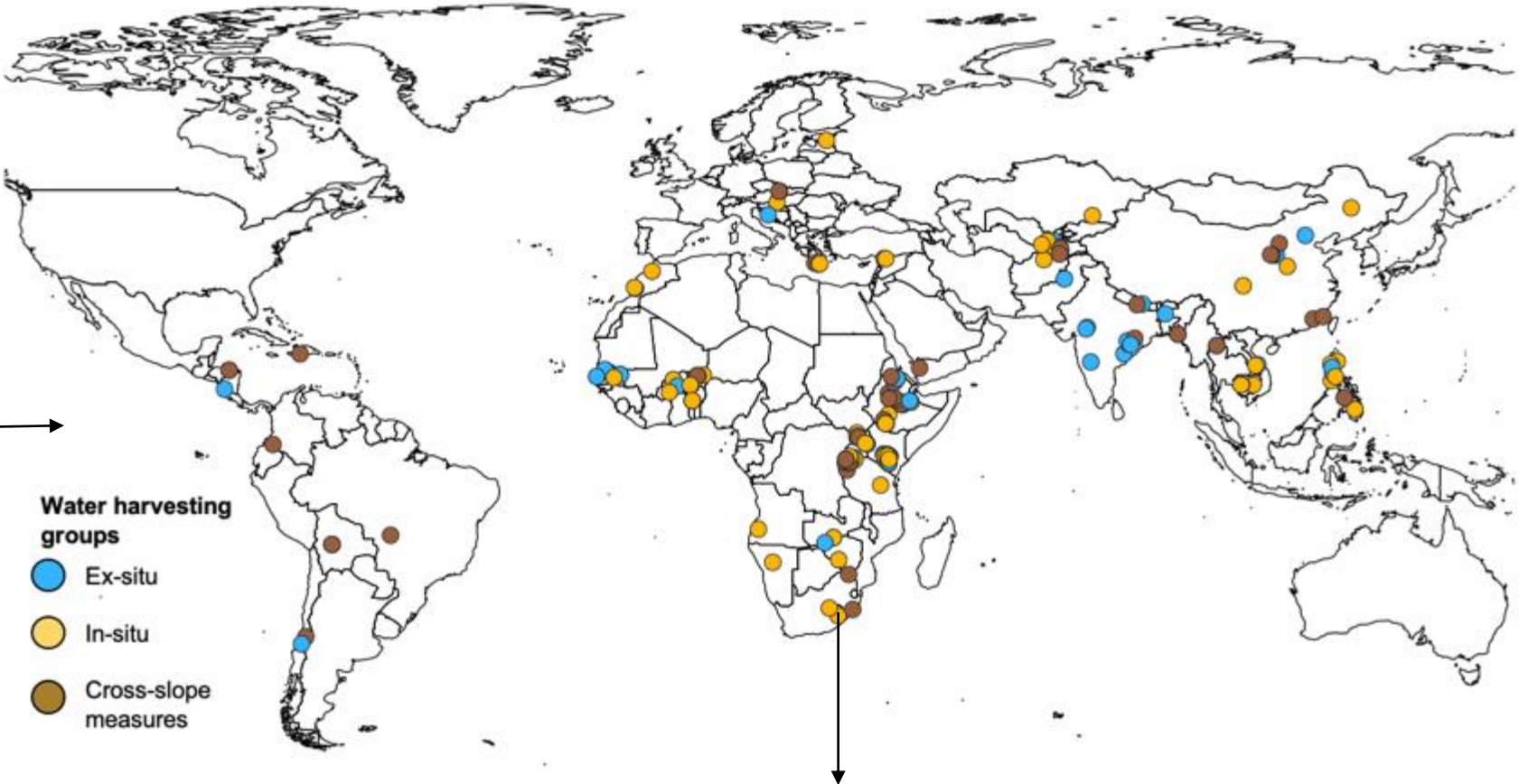
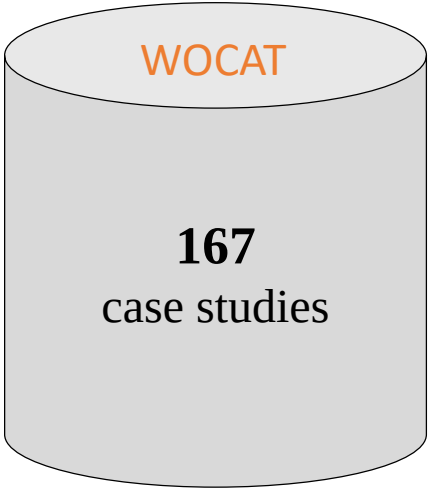
Socio-economic impacts					
Crop production	decreased			✓	increased
irrigation water availability	decreased			✓	increased
farm income	decreased			✓	increased

Quantity before SLM: 60
Quantity after SLM: 80
Technology reduce flood and lower lands doesnt effect by floods

Quantity before SLM: 20
Quantity after SLM: 25
Increasing under ground water by applying technology

Quantity before SLM: 0
Quantity after SLM: 50
Cultivation of cash crop in the terraces (farm income from uplands)

Evidence-based



Socio-economic impacts						
Crop production	decreased			✓		increased
irrigation water availability	decreased			✓		increased
farm income	decreased			✓		increased

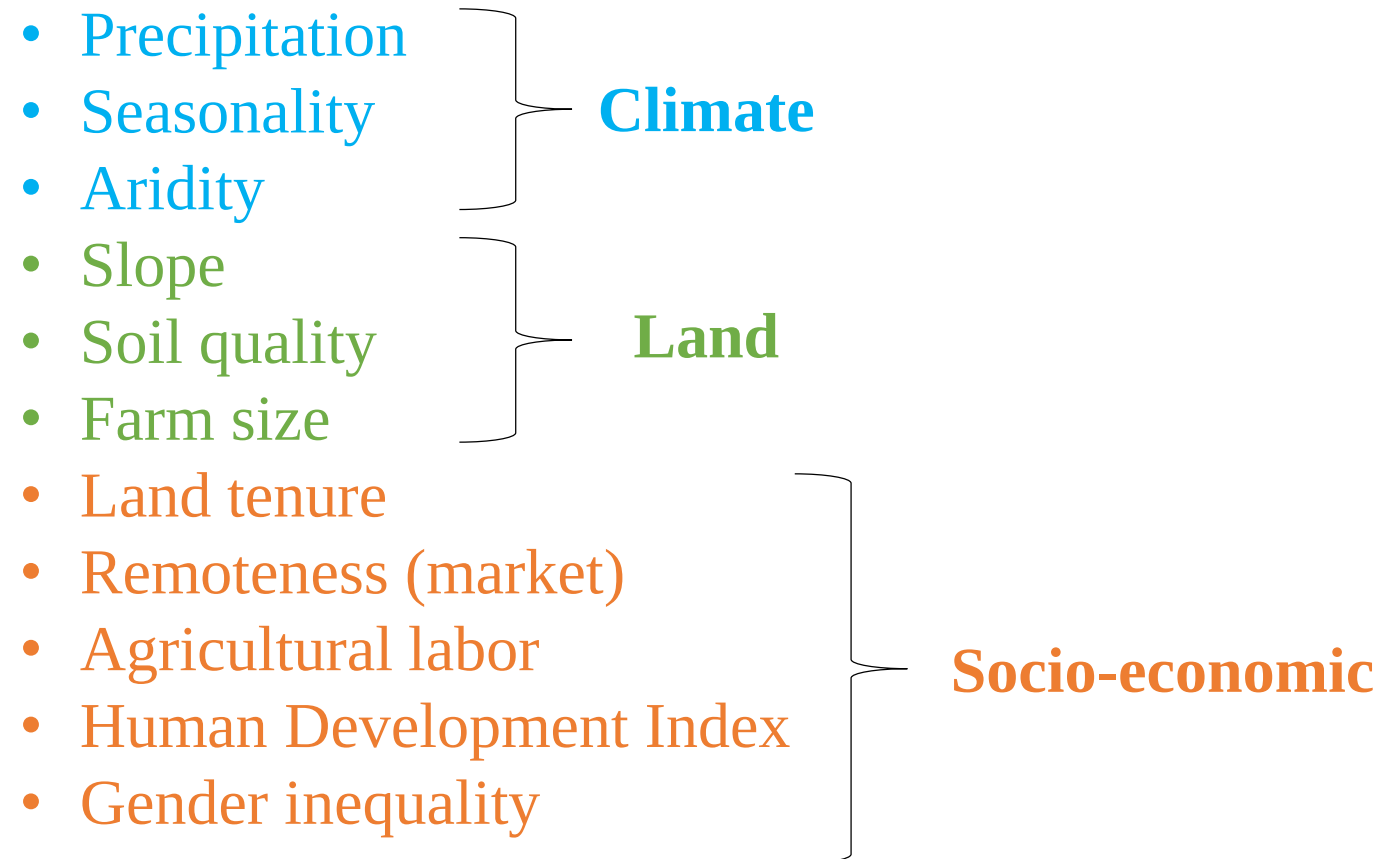
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Quantity after SLM: 25
Increasing under ground water by applying technology
Quantity before SLM: 0
Quantity after SLM: 50
Cultivation of cash crop in the terraces (farm income from uplands)

Assumption:

The increase in crop production of a case study can be replicated in areas with **similar social-ecological conditions**

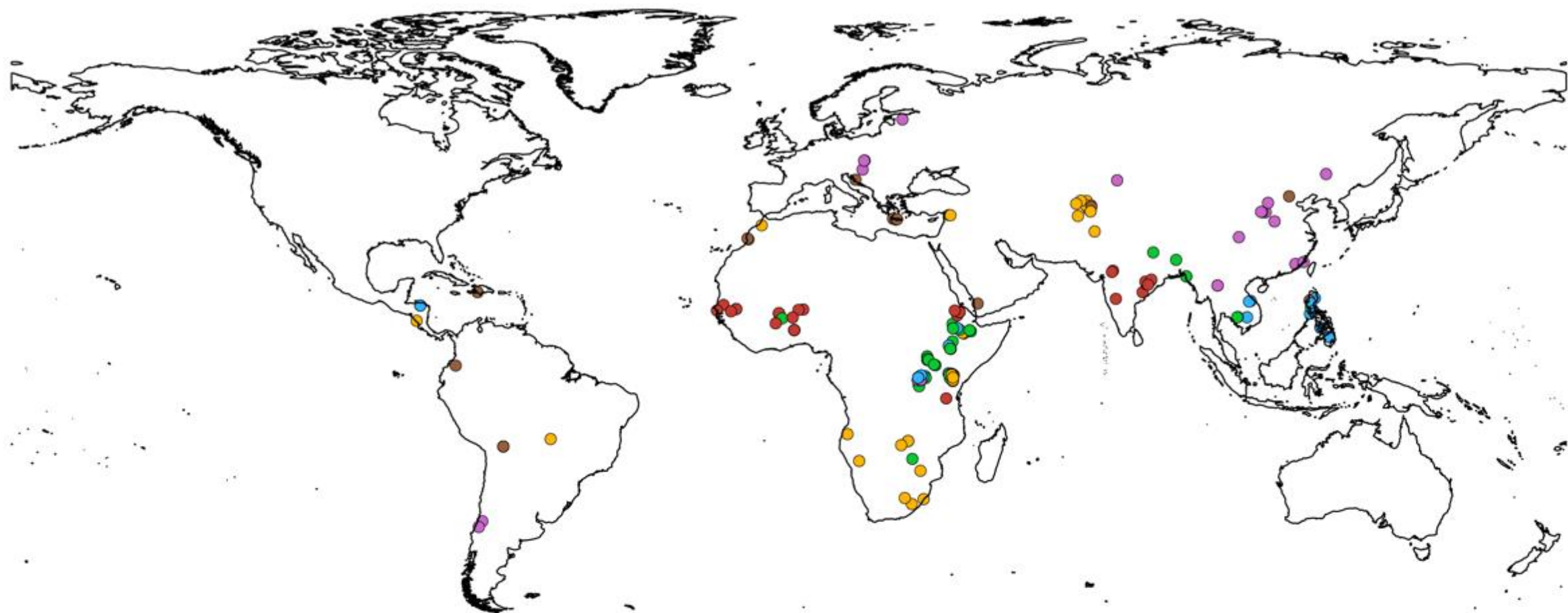
Social-ecological perspective

11 social-ecological datasets

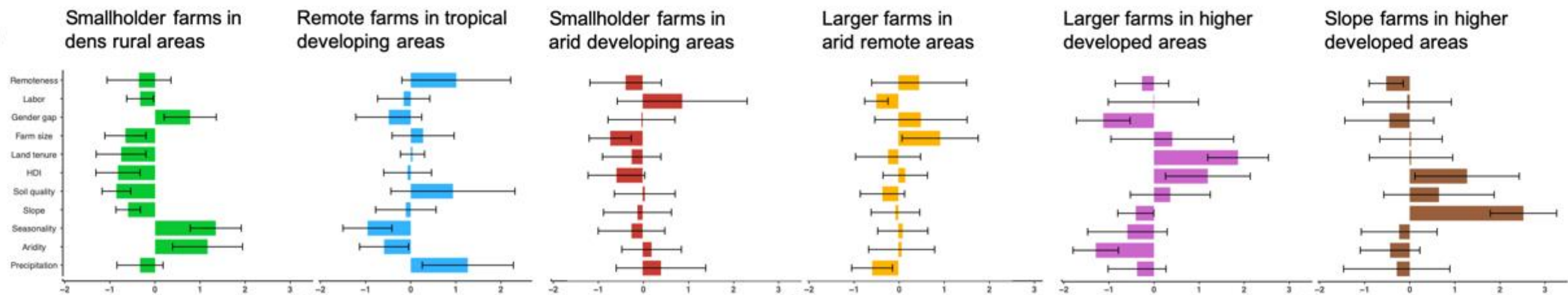


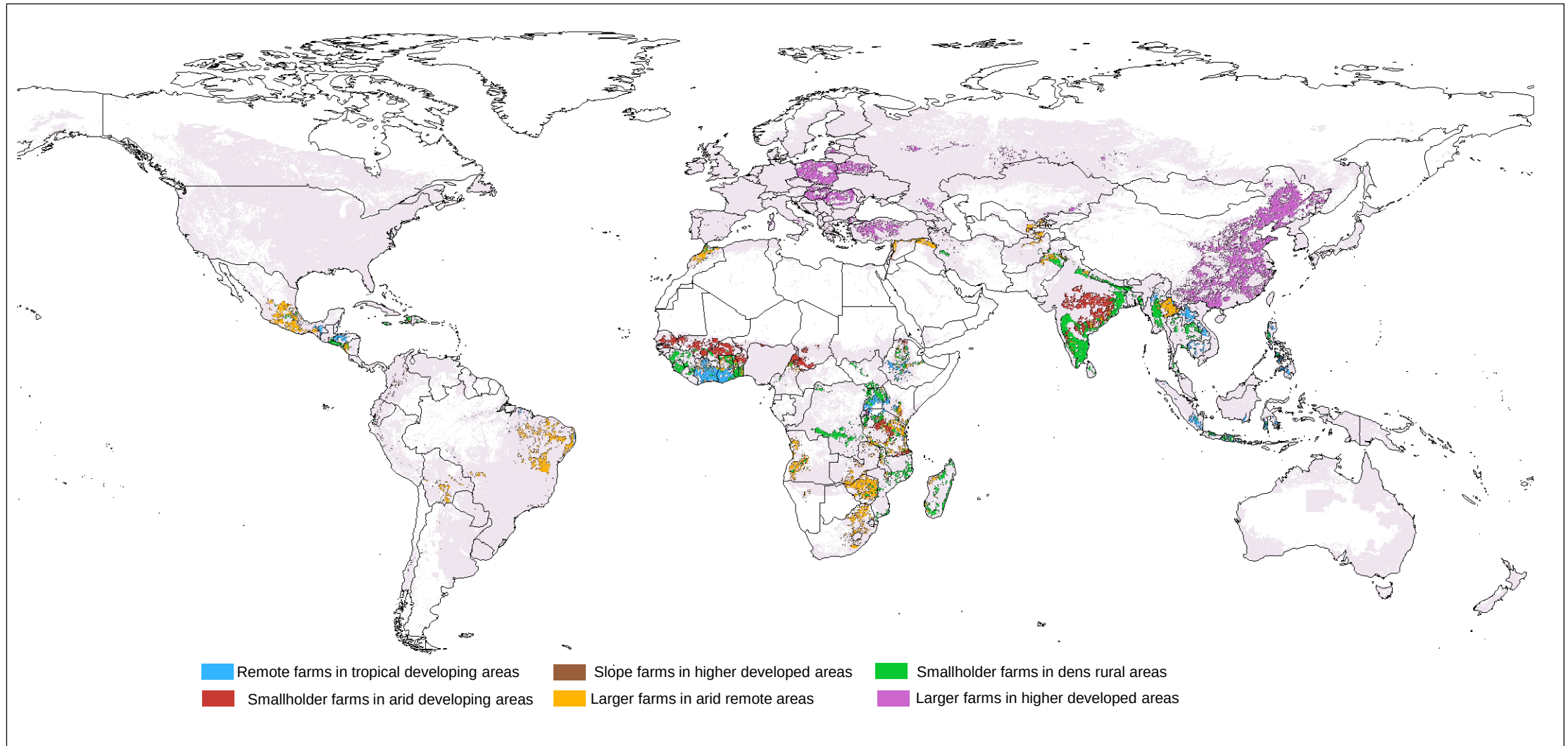
Results

a)



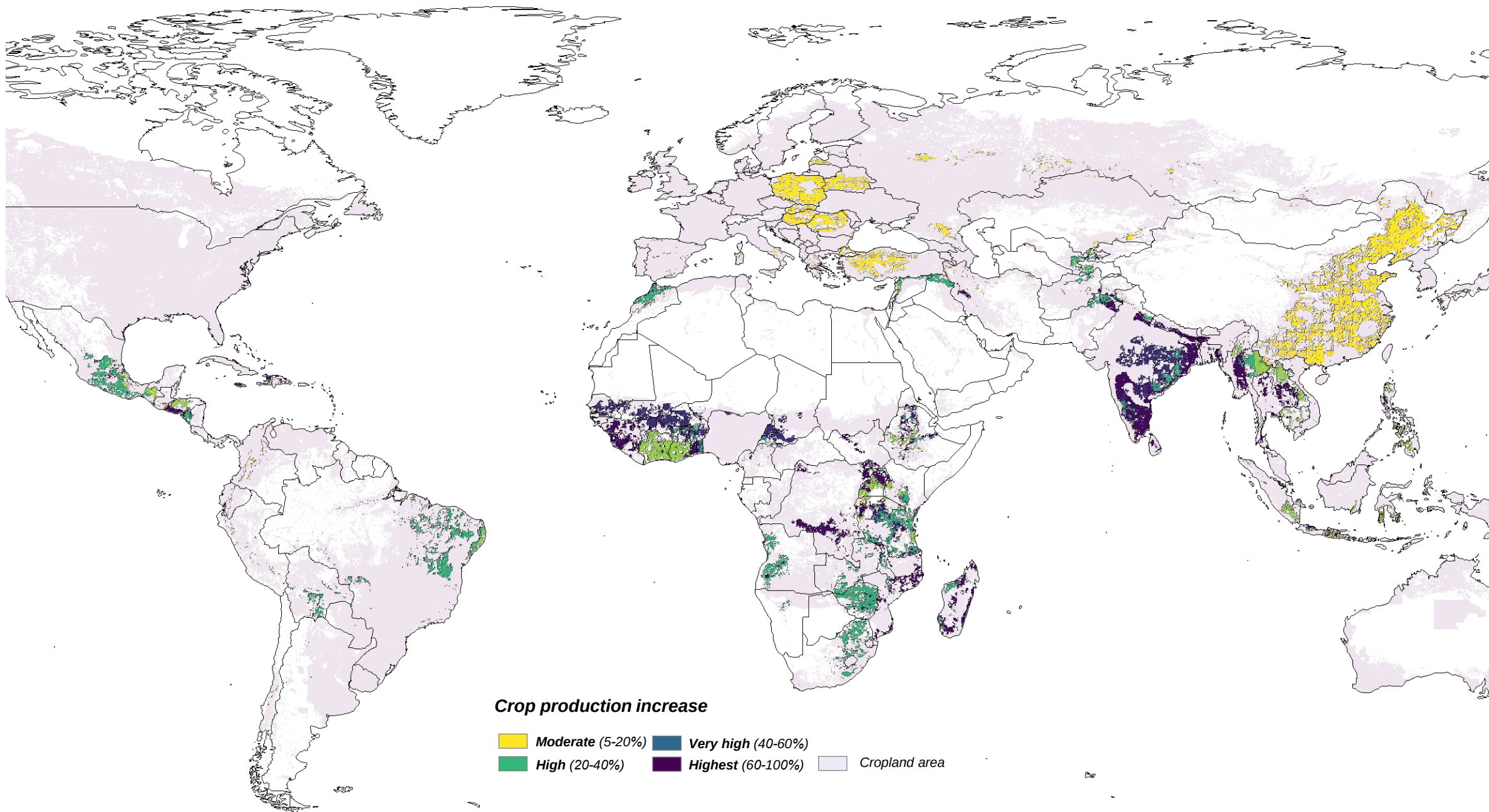
b)



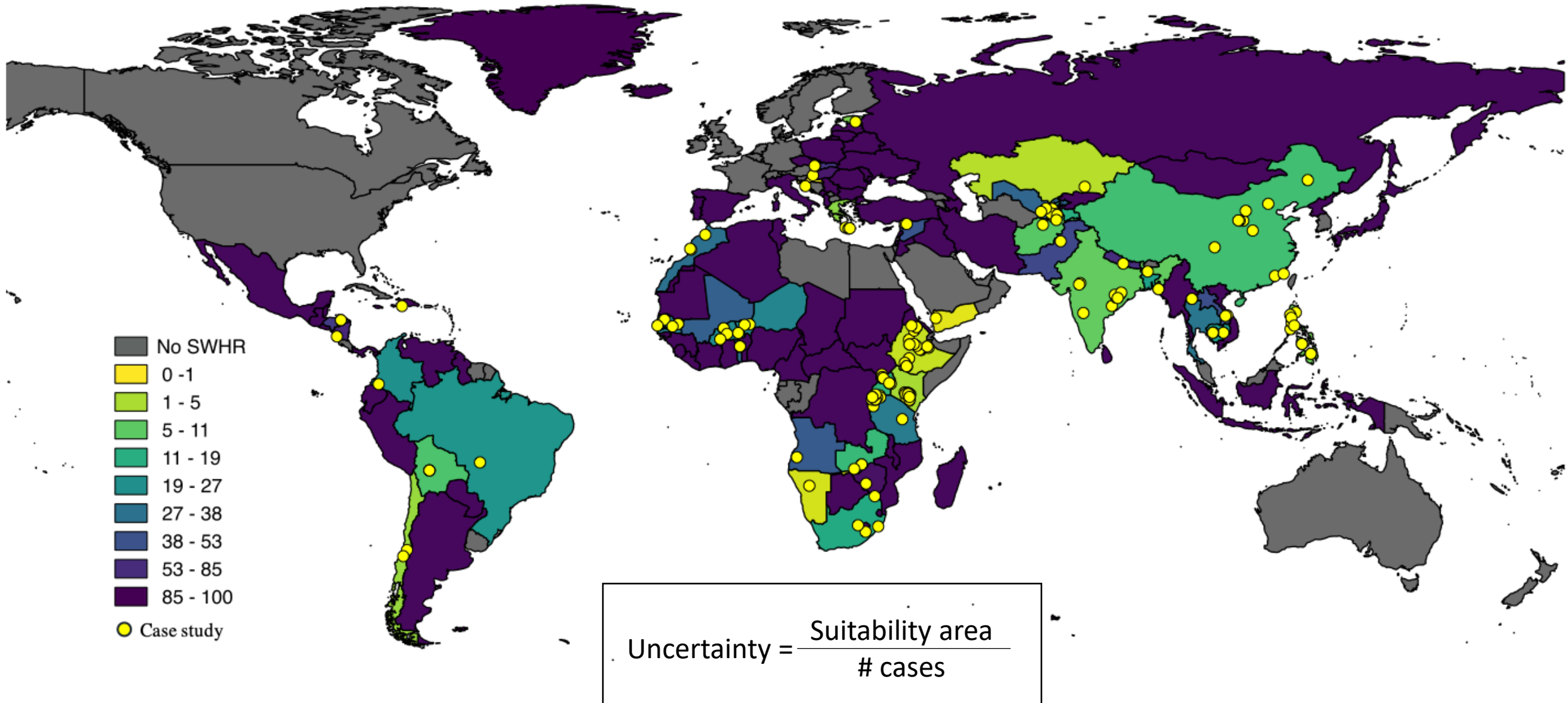


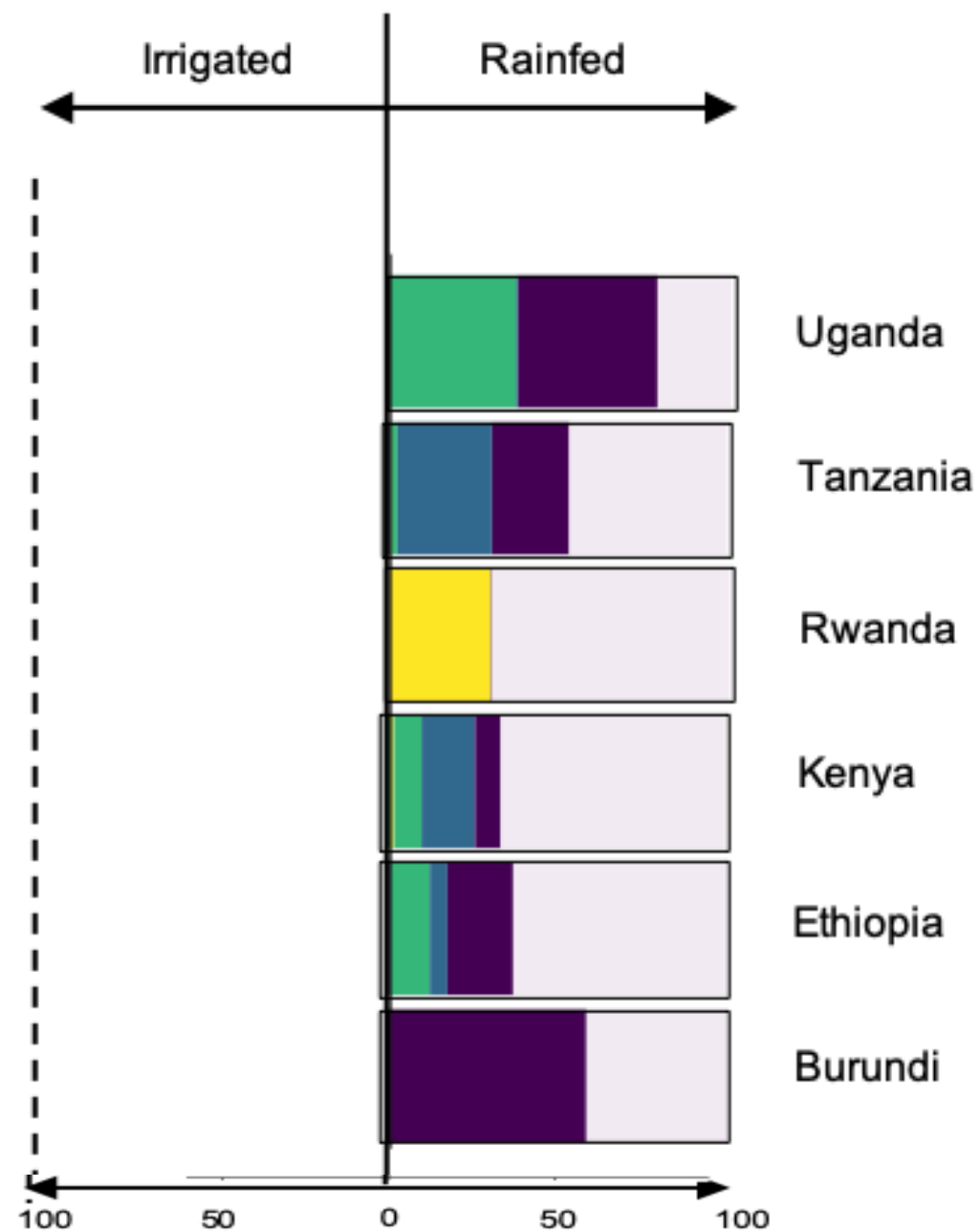
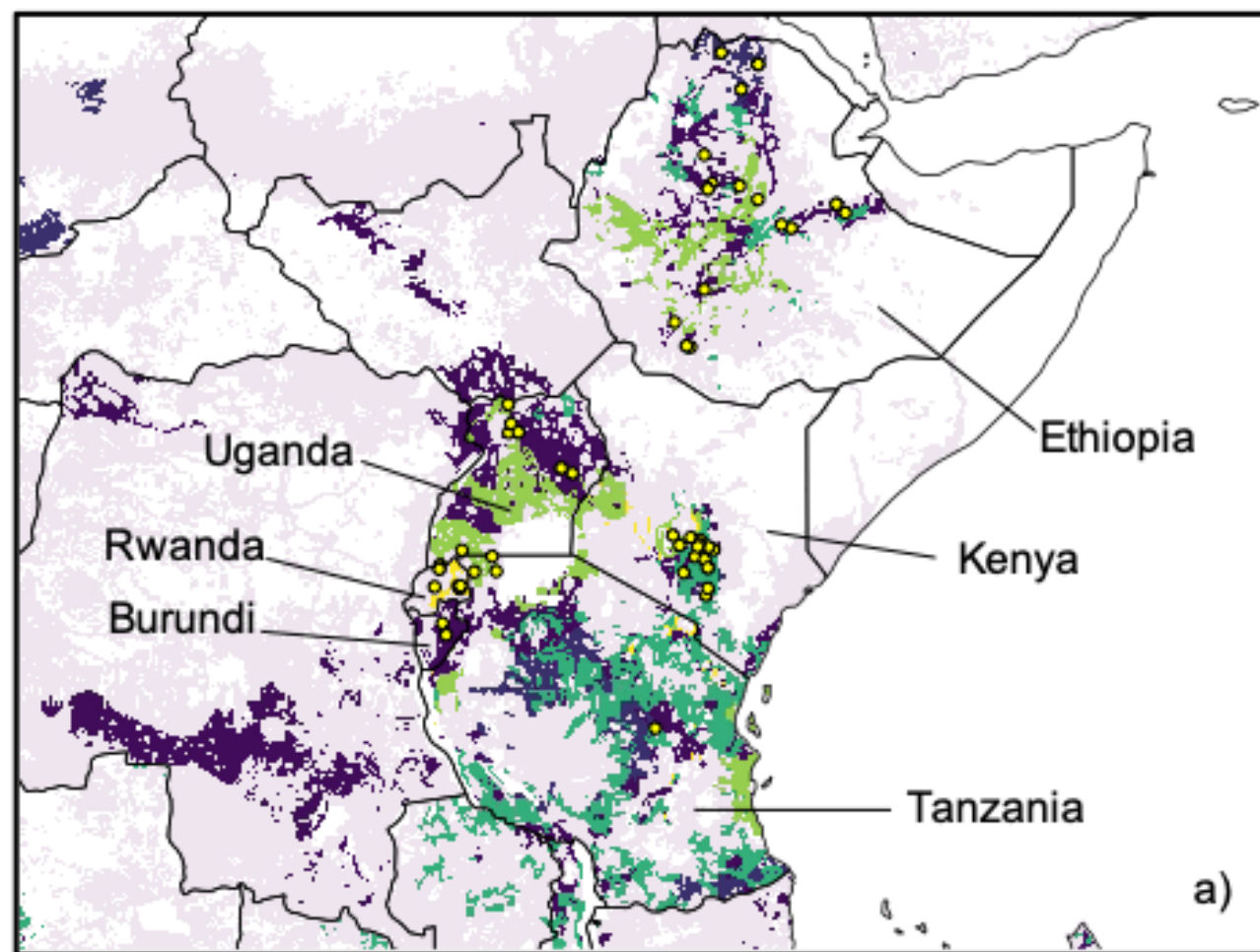
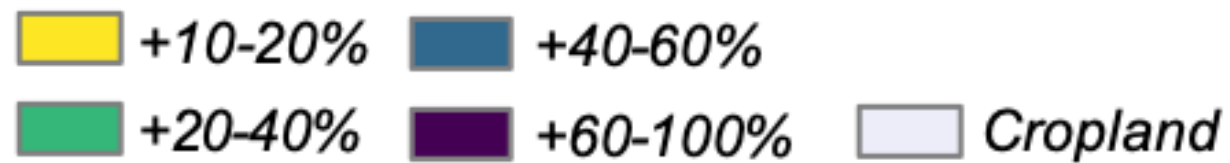
Social-ecological regions for the transferability of WH case studies

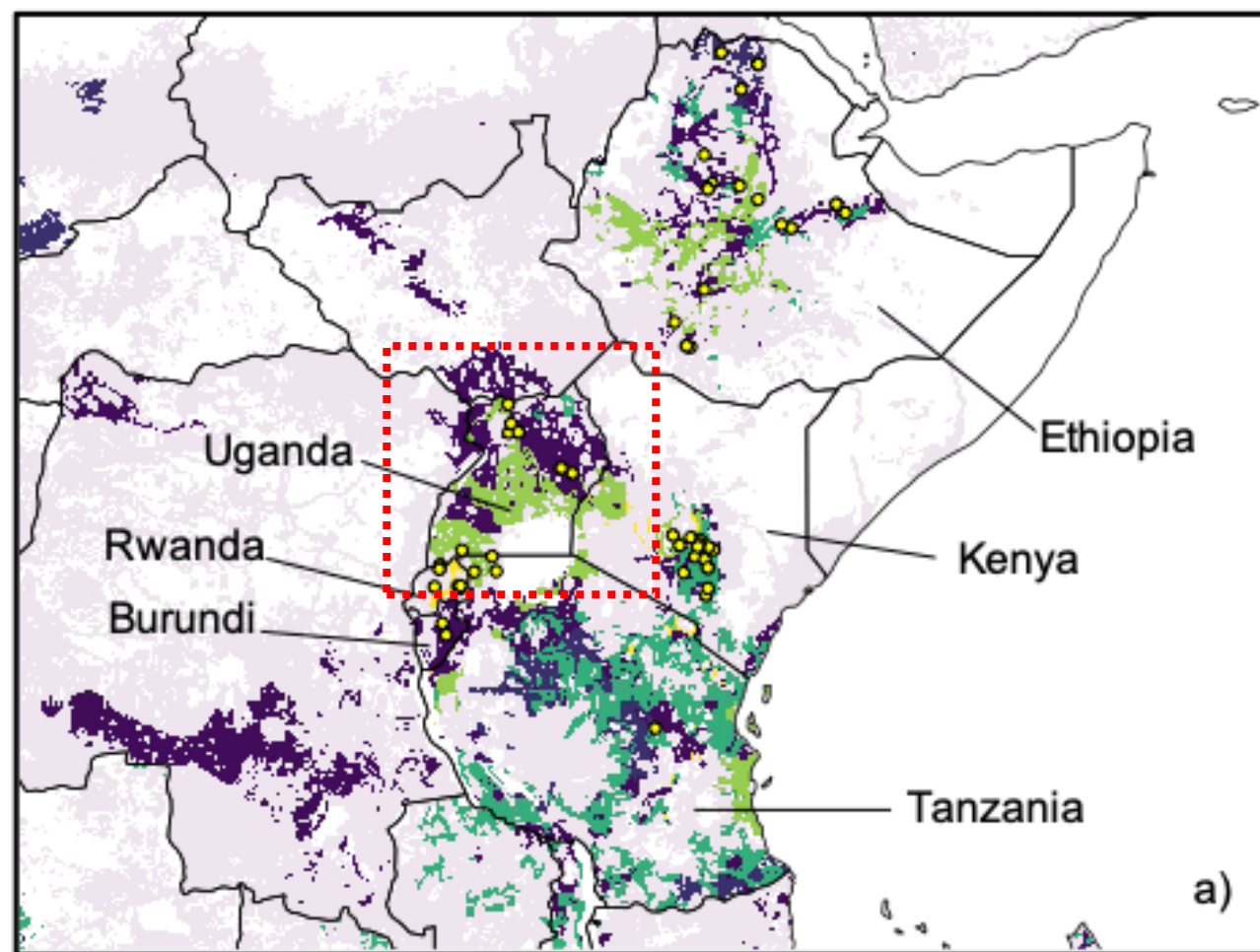
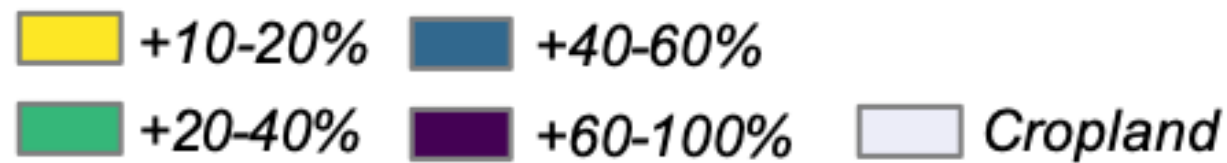
19% of global agricultural land



Uncertainty?



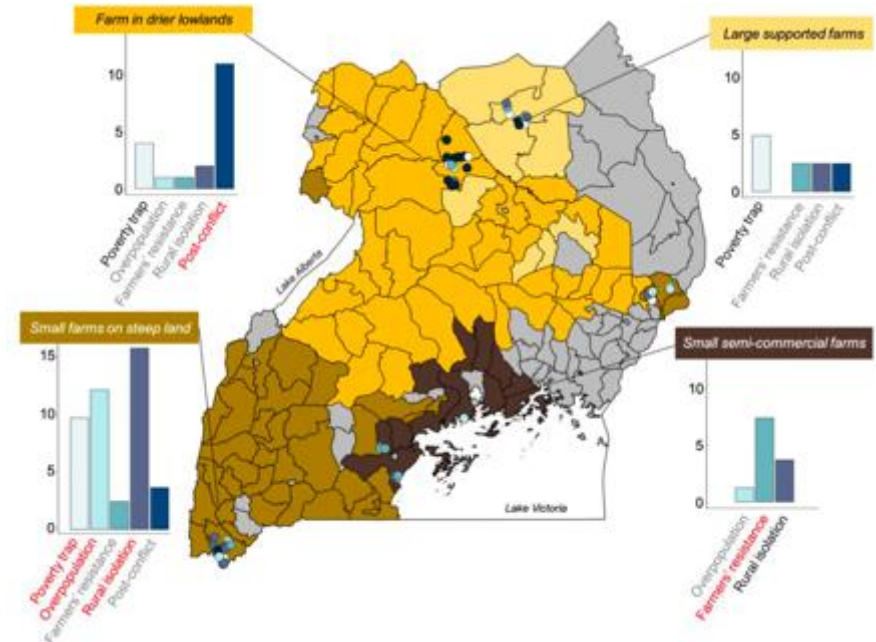




Follow up

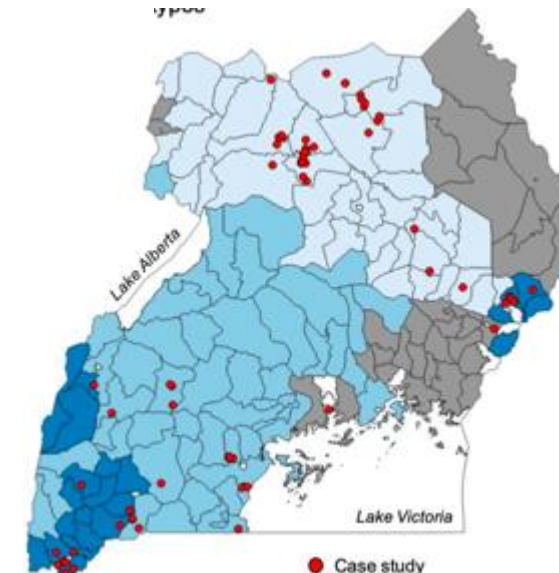
What are the barriers to scaling SLWM in Uganda?

Piemontese, L., R. N. Kamugisha, J. M. B. Tukahirwa, A. Tengberg, S. Pedde, and F. Jaramillo. 2021. Barriers to scaling sustainable land and water management in Uganda: a cross-scale archetype approach. Ecology and Society 26(3):6.
<https://doi.org/10.5751/ES-12531-260306>



What is the investment required to upscale the successful SLWM practices in Uganda?

Piemontese, L., Kamugisha, R.N., Barron, J., Tukahirwa, J.M.B., Harari, N., Jaramillo, F., 2022. Investing in sustainable intensification for smallholders: quantifying large-scale costs and benefits in Uganda. Environ. Res. Lett. <https://doi.org/10.1088/1748-9326/ac5ae0>



Thanks!

luigi.piemontese@unifi.com

Water Harvestin Lab: www.dagri.unifi.it/vp-261-wh-lab.html

WOCAT database: <https://qcat.wocat.net/en/wocat/list/?type=technologies>



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